

Perceived Income Risk and Its Dynamics Across the Income Distribution

1 Introduction

In the New York Fed’s Survey of Consumer Expectations, low earners report wage uncertainty roughly two and a half times that of high earners, and their belief updating is more persistent and responsive to wage shocks. Their realised wage process, measured in matched SIPP earnings panels, shows no comparable gap. The wedge sits on the belief side. This paper documents the empirical pattern, proposes a microfoundation to explain it, and embeds the result in a quantitative model.

Two strands in the labour-market literature suggest where the wedge may come from. The first documents that low-income paychecks display the highest monthly volatility, driven by hours and gig variation (Hardy and Ziliak, 2014; Morduch and Schneider, 2017), and that recession-driven losses fall hardest on low-income workers (Hoynes et al., 2012). Low-income workers therefore observe a noisier income signal than the underlying wage process alone would suggest. The second documents that compensation structures are systematically income-graded: bonus, commission, and performance pay are concentrated at the top (Lemieux et al., 2009), and the shock shape of high-income earnings is predominantly transitory while low-income shocks are persistent (Guvenen et al., 2021). Low-income workers therefore see a smaller share of their transitory shocks arrive in institutionally identifiable form. Both are observable facts of the labour market that should have an income-graded impact on expectations, but have not yet been linked to subjective beliefs about income risk.

I propose a structural microfoundation that ties the strands together. Two features of compensation, both income-graded against the bottom, jointly generate the perceived-risk wedge: paycheck noise that produces slower belief updating and an inflated transitory prior, and a smaller share of identifiable transitory pay that produces a larger response per unit shock at the bottom. The empirics verify this relation: the bottom tercile treats a much larger share of each wage shock as permanent than the middle and top, and carries a prior on transitory-variance that is far above what their realised wage data implies; the middle and top extrapolate at about half this rate and hold priors close to their realised values.

I conjecture that these wedges will have the following macroeconomic implications: cross-sectional dispersion in precautionary saving with GE effects, and MPCs out of wage news that a FIRE calibration cannot reproduce, with the steepest gradient at the bottom of the income distribution. In this memo I propose an Aiyagari implementation of the calibrated belief structure, and will want to compare its predictions against the FIRE benchmark.

2 Data

Sample period 2013–2023 for SIPP, 2013–2025 for SCE. Both are monthly within-respondent panels, following each individual across consecutive months in the same survey. I Mincer-residualise on standard demographics and year-month FE, restrict to same-job/same-employer spells, and require at least 6 consecutive monthly observations per respondent. Respondents are sorted into three income groups by median annual household income: Bot (<\$50k), Mid (\$50k–\$100k), Top ($\geq$ \$100k).

SCE. NY Fed Survey of Consumer Expectations. Per-respondent variance of next-12-month earnings growth, computed from the 10-bin density forecast on the current job. The density question conditions on “*same job, hours, employer*”; I drop each respondent’s observations from the first wave they report a job change. Perceived variance trimmed at 1%/99%.

SIPP. Survey of Income and Program Participation. Monthly wage histories within stayer spells. I difference consecutive observations within each spell to recover monthly wage changes, and trim the squared changes at the 1%/99% window.

3 Perceived risk is higher and more persistent at the bottom

Perceived beliefs are higher at the bottom

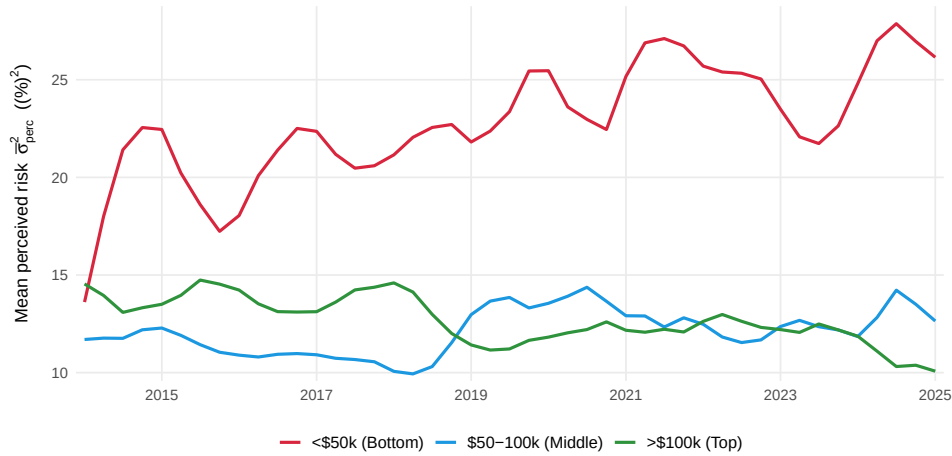


Figure 1: Group-mean $\bar{\sigma}_{\text{perc}}^2$ in $(\%)^2$, quarterly with 4-quarter trailing MA. Matched-stayer SCE panel, Mincer-residualised on demographics and month-year FE.

SCE perceived variance is steeply income-graded throughout the sample period. The weighted tercile means of $\bar{\sigma}_{\text{perc}}^2$ come in at Bot **25.1**, Mid 12.0, Top 9.9, so Bot’s perceived variance runs roughly **2.5** $\times$ Top’s. The gradient is stable across the sample, with Bot showing the largest persistent deviations, possibly reflecting greater exposure to the business cycle.

Perceived beliefs are also more persistent at the bottom

To explore the persistence, I use a within-respondent AR(1) on $\bar{\sigma}_{\text{perc}}^2$, estimated via Blundell-Bond system GMM. I use BB throughout the paper following Ma et al. (2020) on managers' forecast errors, who work in the same short-panel-with-Nickell-bias setting:

Tercile	Monthly		Quarterly	
	$\hat{\rho}_g$	SE	$\hat{\rho}_g$	SE
Bot	+0.264***	0.037	+0.590***	0.050
Mid	+0.144***	0.033	+0.536***	0.077
Top	+0.146***	0.030	+0.395***	0.071
Wald ($B = T$)	$p = 0.013$		$p = 0.024$	

Table 1: Within-respondent AR(1) persistence of perceived variance. Two-way clustered SEs. Instruments: $\bar{\sigma}_{\text{perc}}^2$ at lags 3–4 and S at lags 1–3, BB level moments.

Bot's perceived variance is roughly twice as persistent as Top's, and the gap is statistically significant at both monthly and quarterly cadences.

So perception is income-graded in both level and persistence. The standard practice in HA macro is to calibrate σ^2 from realised wage data and assume perceived equals realised. *That conflicts with both moments.*

4 The realised wage process cannot explain the wedge

Wage process as the lens

What might generate the perceived-risk gradient and its dynamics? Standard wage theory points to the realised income process, and Figure 2 confirms its relevance: $\bar{\sigma}_{\text{perc}}^2$ co-moves strongly with the absolute deviation of aggregate wage growth from its sample mean.¹ I hypothesize is that surprises to expected income growth drive the observed dynamics in perceived risk.

¹I use absolute deviations since any realised shock should increase the perceived uncertainty of one's income process. Using squared deviations gives a very similar plot. I later split the shocks into positive and negative components to test for asymmetric effects on perceived income risk.

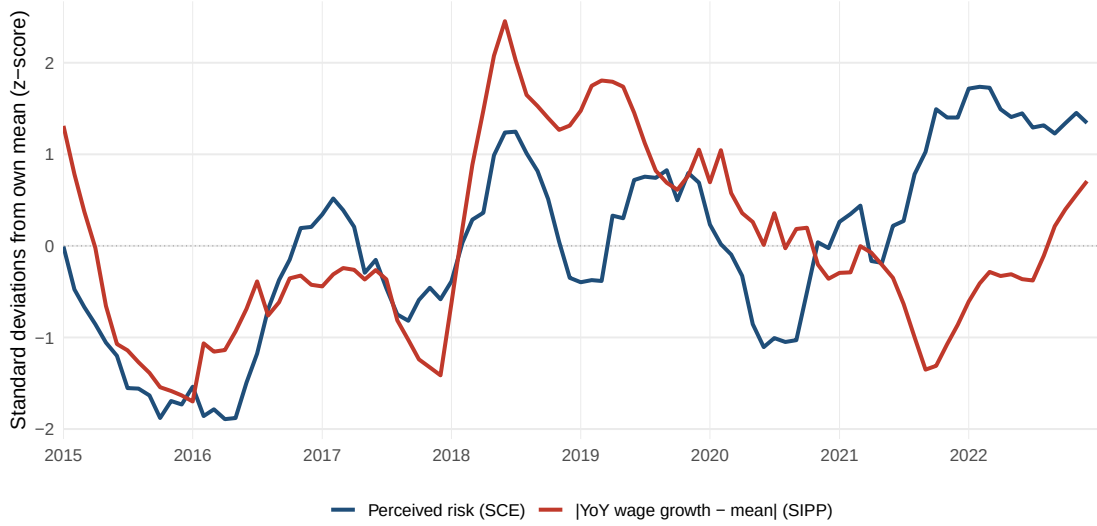


Figure 2: SCE perceived risk and the absolute deviation of SIPP aggregate wage growth from its sample mean pooled across all income groups, 2015–2022.

Wage model

Standard permanent-transitory decomposition (Meghir and Pistaferri, 2004):

$$w_{i,t} = z_{i,t} + e_{i,t}, \quad e_{i,t} = p_{i,t} + \theta_{i,t}, \quad p_{i,t} = p_{i,t-1} + \psi_{i,t}.$$

$z_{i,t}$ is the predictable Mincerian path. $\psi_{i,t}$ is the permanent shock to earning power, iid with variance σ_ψ^2 , accumulating into $p_{i,t}$ via a random walk (inflation adjustment, merit raise, COLA). $\theta_{i,t}$ is the transitory shock, iid with variance σ_θ^2 (discretionary bonus, profit-sharing).

Matching SCE to the model

The SCE density question asks about $\Delta w_{i,t+1}$ *conditional on same job, hours, employer*. This conditioning pins $\Delta z = 0$, so the elicited variance is the variance of the stochastic component alone:

$$\bar{\sigma}_{\text{perc},i,t}^2 \equiv \text{Var}_{i,t}(\Delta w_{i,t+1} \mid \Delta z = 0) = \text{Var}_{i,t}(\Delta e_{i,t+1}).$$

On SIPP, I residualise $\log w$ on Mincer controls and difference to get Δe directly. Both sides now measure $\text{Var}(\Delta e)$, so the question is whether the subjective measurement tracks the realised one. Answering it requires decomposing the realised side into its structural primitives to see where they might differ.

Standard MP (2004) identification

Given the wage model above, the one-period change in the stochastic component is $\Delta e_t = \psi_t + \theta_t - \theta_{t-1}$. From the sample I estimate two moments per tercile:

$$\bar{S}_m \equiv \mathbb{E}[(\Delta e)^2] = \sigma_\psi^2 + 2\sigma_\theta^2, \quad \text{Cov}(\Delta e_t, \Delta e_{t-1}) = -\sigma_\theta^2.$$

The lag-1 autocovariance pins σ_θ^2 , and the remaining variance pins σ_ψ^2 . Two moments and two unknowns leave the wage primitives exactly identified per tercile.

Tercile	σ_θ^2 (transitory)	σ_ψ^2 (permanent)	$\bar{S}_m$
Bot	2.52	11.92	16.96
Mid	4.42	11.61	20.45
Top	5.83	11.23	22.89

Table 2: SIPP MP identification per tercile, monthly.

Two facts: permanent variance is roughly flat across terciles (≈ 11.5); transitory variance rises mildly with income, plausibly through a bonus-pay channel at the Top. Neither component reproduces the steep gradient in perceived variance. If anything, the realised gradient runs in the opposite direction.

The FIRE benchmark

With the estimated wage-process parameters, I now compare what the model says the SCE response should look like with what respondents actually report. Rolling the monthly process forward 12 months under iid shocks gives (*simple derivation in Appendix B*)

$$\text{Var}(e_{t+12} - e_t) = \underbrace{\sum_{k=1}^{12} \sigma_\psi^2}_{\text{permanent}} + \underbrace{2\sigma_\theta^2}_{\text{transitory}} = 12\sigma_\psi^2 + 2\sigma_\theta^2, \quad (1)$$

which is what a FIRE agent (who knows the parameters) would report. Permanent variance scales linearly with horizon (12 iid permanent innovations stack up); transitory variance is horizon-invariant (only the endpoints matter for an iid blip). Plug in SIPP values:

Tercile	FIRE benchmark ($12\sigma_\psi^2 + 2\sigma_\theta^2$)	SCE observed $\bar{\sigma}_{\text{perc}}^2$	ratio
Bot	148.1	25.1	5.9×
Mid	148.2	12.0	12.4×
Top	146.4	9.9	14.8×

Table 3: FIRE-implied 12-month variance vs SCE-reported, in $(\%)^2$.

The FIRE benchmark deviates from the SCE perceived variance on three counts: a flat cross-section is predicted versus a steep gradient observed, the absolute level is off by 6–15× in variance, and the model implies no time-variation while SCE perception is clearly dynamic (Section 2).

I therefore need a belief rule that i) allows perceived variance to differ from realised, ii) evolve with realised shocks, and iii) differ across terciles.

5 A belief rule for the FIRE deviations

What a belief rule needs to deliver

The three FIRE failures each require a structural element. First, the rule must allow dynamic updating, since SCE perceived variance is persistent and time-varying; allowing the update speed to differ across terciles also delivers the persistence gradient. Second, the rule must allow realised wage shocks to enter the update. The realised wage process is roughly equal across terciles, so the cross-tercile level wedge cannot come from differential signals; it must come from differential extrapolation of the same signal, which is also what lets perceived risk co-move with realised risk (Figure 2). Third, the rule must allow a baseline belief about transitory risk that the agent never updates: Top’s reported SCE level sits below what realised transitory shocks alone would produce, so even pushing extrapolation to zero cannot reach it. The static baseline must be free to sit below the wage-implied truth, otherwise the model is forced into negative attribution.

The agent’s problem

To microfound the FIRE deviations above I appeal to a Kalman-style learning rule: agents over-weight recent signals when updating beliefs about wage risk. The agent maintains a belief about her permanent wage variance but cannot see permanent and transitory shocks separately; each period she observes one squared wage change $(\Delta e_t)^2$, which mixes the two by construction ($\mathbb{E}[(\Delta e)^2] = \sigma_\psi^2 + 2\sigma_\theta^2$). She must therefore attribute each surprise: a portion to a permanent shift in earning power (σ_ψ^2), the rest to a transitory blip (σ_θ^2).

Modelling choice

The agent has two unknown variances, but SCE delivers only one perceived-variance series per tercile. That is enough to pin one dynamic, not two. Therefore, I let the permanent belief $\hat{\sigma}_{\psi,t}^2$ evolve and treat the transitory belief as a static prior:

$$\sigma_{\theta,\text{perc}}^2 = \text{constant per tercile, fixed across time.}$$

The permanent component is the welfare-relevant variance (permanent shocks change lifetime earning power; transitory ones fade quickly), so the dynamics there are what matters. The cost of this asymmetry is that any belief-side mis-attribution, including in the transitory belief, gets absorbed by λ below.

The updating rule

I model the permanent-variance belief with an exponentially-weighted moving average (EWMA) on past squared shocks, with the attribution share built in as a separate weight:

$$\hat{\sigma}_{\psi,t}^2 = \underbrace{(1 - \mu)}_{\text{persistence}} \hat{\sigma}_{\psi,t-1}^2 + \underbrace{\mu}_{\text{update weight}} \underbrace{\lambda}_{\text{attribution share}} (\Delta e_{t-1})^2. \quad (2)$$

This is the limiting case of GARCH(1,1) with parameters summing to one, or similarly the steady-state Kalman filter for an unknown variance under a random-walk prior. The two parameters have clean interpretations:

- μ is the speed at which beliefs update. Higher μ means each period's news shifts the belief more; lower μ means beliefs are sticky and the past dominates.
- λ is the fraction of each squared surprise the agent treats as informative about the permanent component. $\lambda = 0$ means she reads every shock as fully transitory and never updates her permanent belief from realised data; higher λ means she treats more of each shock as informative.

Hence, taking the Eq. (1) FIRE decomposition $\bar{\sigma}_{\text{perc}}^2 = 12\sigma_{\psi}^2 + 2\sigma_{\theta}^2$ and swapping the true parameters for the agent's beliefs gives the level she reports each period:

$$\bar{\sigma}_{\text{perc},t}^2 = 12\hat{\sigma}_{\psi,t}^2 + 2\sigma_{\theta,\text{perc}}^2. \quad (3)$$

FIRE in the belief rule

Strict FIRE corresponds to $\mu = 0$ in Eq. (2) with both beliefs at the truth ($\hat{\sigma}_{\psi,0}^2 = \sigma_{\psi}^2$ and $\sigma_{\theta,\text{perc}}^2 = \sigma_{\theta}^2$): reported variance is then constant at $12\sigma_{\psi}^2 + 2\sigma_{\theta}^2$. That corner is ruled out by the (non-static) dynamics displayed in Section 3.

Within the learning family ($\mu > 0$), FIRE can only survive as a *long-run benchmark*: the agent's belief recovers the truth in steady-state expectation when

$$\lambda^* = \sigma_{\psi}^2 / \bar{S}_m.$$

The transitory side has its own FIRE condition: $\sigma_{\theta,\text{perc}}^2 = \sigma_{\theta}^2$. Combined, long-run FIRE requires both $\lambda = \lambda^*$ and $\sigma_{\theta,\text{perc}}^2 = \sigma_{\theta}^2$ to hold simultaneously at every tercile. Section 7 will show neither does.

6 Estimation strategy

The belief rule of Section 5 is a recursion on the unobservable belief state $\hat{\sigma}_{\psi}^2$. Combining the updating rule (2) with the level identity (3) converts it into an AR(1) in observable perceived variance (*derivation in Appendix A*):

$$\bar{\sigma}_{\text{perc},i,t}^2 = \underbrace{2\mu_g \sigma_{\theta,\text{perc},g}^2}_{\beta_{\text{cons}}} + \underbrace{(1 - \mu_g)}_{\beta_{\text{lag}}} \bar{\sigma}_{\text{perc},i,t-1}^2 + \underbrace{12\mu_g \lambda_g}_{\beta_{\text{shock}}} (\Delta e_{i,t-1})^2 + \varepsilon_{i,t}. \quad (4)$$

The three reduced-form coefficients map invertibly to the three structural parameters of Section 5:

$$\mu_g = 1 - \hat{\beta}_{\text{lag}}, \quad \sigma_{\theta,\text{perc},g}^2 = \frac{\hat{\beta}_{\text{cons}}}{2\mu_g}, \quad \lambda_g = \frac{\hat{\beta}_{\text{shock}}}{12\mu_g}.$$

Each coefficient picks up a distinct empirical pattern: β_{lag} from the within-respondent persis-

tence of perceived variance; β_{cons} from the cross-tercile level after netting out the news loading; β_{shock} from how much perceived variance moves with the lagged realised wage shock.

The shock regressor

The belief rule (2) calls for the realised squared wage change $(\Delta e_{i,t-1})^2$ of the specific agent reporting the belief. SCE has no own wage histories, so I substitute the SIPP cell-mean over workers in the same cell $c = (\text{tercile} \times \text{Census division} \times \text{month})$, following Guerreiro (2023). Geography enters because workers in the same local labour market share components of wage news (industry mix, local demand), which agents plausibly absorb through coworkers and local media.² Census divisions balance power and resolution: state cells would be too sparse, while a national pool would discard regional heterogeneity. I smooth the cell series with a 3-month trailing moving average to reduce sampling noise.

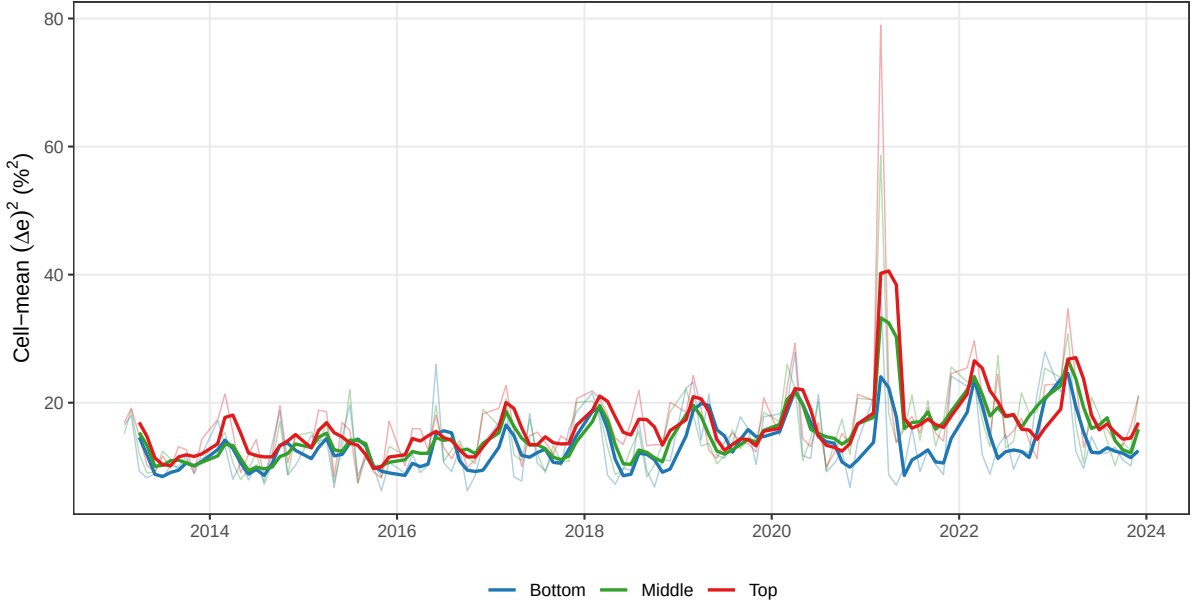


Figure 3: Cell-mean $(\Delta e_{i,t})^2$ by income tercile, national monthly, 2013–2024, in $(\%)^2$. Thin faint lines are the raw monthly cell-mean; thick solid lines are the 3-month trailing moving average actually used as the regressor in Eq. (4).

Top shows the largest amplitude in $(\Delta e)^2$, consistent with the higher σ_θ^2 from Section 4 and the documented concentration of discretionary pay at the top of the income distribution (Güvenen et al., 2021; Halvorsen et al., 2024). Since Bot’s wage shocks are no larger than Top’s, any cross-tercile gradient in $\hat{\lambda}$ must reflect a behavioural channel: Bot’s belief would need to respond more strongly to each wage shock than Top’s.

²See i.e. Kuchler and Zafar (2019) on the salience of personal experiences in shaping aggregate expectations.

7 Headline result

Joint GMM on (4) delivers the conjectured monotone pattern across the terciles, with both belief channels (attribution share and transitory prior) most distorted at the bottom.

Tercile	σ_θ^2 (SIPP truth)	$\sigma_{\theta,\text{perc},g}^2$ (GMM)	λ_g (GMM)	μ_g (GMM)	Hansen J p
Bot	2.52	9.75 (1.71)	+0.077 (0.030)	0.63 (0.06)	0.20
Mid	4.42	2.94 (0.93)	+0.046 (0.015)	0.91 (0.09)	0.23
Top	5.83	4.12 (0.82)	+0.026 (0.013)	0.95 (0.06)	0.24
Wald ($B = T$)	—	$p = \mathbf{0.003}$	$p = 0.12$	$p = \mathbf{0.0001}$	—

Table 4: Joint BB system GMM on (4). Two-way clustered SEs. Instruments: $\bar{\sigma}_{\text{perc}}^2$ at lags 3–4 and S at lags 1–3, BB level moments (same as in Table 1). First column (σ_θ^2 SIPP truth) taken from Table 2.

Robust specification

Robustness checks across different trimming, alternative lag depths, varying income terciles, and a COVID-excluded subsample leave the cross-tercile gradients largely intact (need to add to appendix). The Hansen J test of over-identifying restrictions does not reject at any tercile, giving no reason to suspect a faulty specification.

Attribution share λ

All three terciles read part of each wage shock as a signal about permanent earning power rather than just a transitory blip, with Bot’s permanent share substantially larger than Mid’s and Top’s. *Implication.* A common wage signal generates disproportionately larger revisions of the permanent-variance belief at the bottom. This is the income-graded transmission channel from realised shocks to perceived risk.

Transitory prior $\sigma_{\theta,\text{perc}}^2$

Bot carries a transitory prior far above her wage-data benchmark; Mid and Top hold priors close to their own. The bottom is the only tercile whose baseline belief is materially detached from its realised wage process. *Implication.* Bot’s inflated prior is the dominant driver of the absolute level of her perceived variance, separate from the news-loading channel.

Update speed μ

Bot updates her belief substantially slower than Mid or Top, with the cross-tercile difference highly significant. The implied AR(1) persistences match the standalone Table 1 persistence estimates, validating the joint specification against the reduced-form benchmark. *Implication.* Bot’s perceived risk reflects older information than Top’s. Sticky beliefs amplify the persistence of any wage news at the bottom of the distribution.

8 Microfounding where the wedges come from

The headline regression delivers three behavioural parameters per tercile $(\mu_g, \lambda_g, \sigma_{\theta, \text{perc}, g}^2)$ with no structural link between them. I propose two primitives, both grounded in observable compensation structures, that together generate the three wedges; supporting references appear in the footnotes.

Two primitives

The agent's paycheck is noisy. The agent does not see her wage process directly. Each month she sees her take-home pay,

$$\Delta y_{i,t} = \Delta e_{i,t} + \nu_{i,t}, \quad \nu_{i,t} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \omega_g^2), \quad (5)$$

where ν is the part of monthly income unrelated to underlying earning power: scheduled-hours volatility and irregular tip income. Bot's labour market generates more of all of these than Top's, so ω_g^2 is income-graded and largest at the bottom.³ The wage primitives $(\sigma_\psi^2, \sigma_\theta^2)$ remain flat across g as Section 4 showed, meaning that all cross-tercile heterogeneity in the paycheck signal Δy_t sits in ω_g^2 .

Transitory pay is heterogeneously labelled. High-income compensation includes visibly-transitory components such as annual bonuses and sales commissions. The same logic covers exposure to macro events: Top's aggregate-shock exposure tends to arrive in classifiable form (furloughs with return dates, cyclical hours cuts), while Bot's arrives in more ambiguous form (sector closures, gig disappearance).⁴ Low-income compensation is otherwise hours times wage plus tips, with no flag separating the transitory portion. Let $\rho_g \in [0, 1]$ be the fraction of transitory pay that arrives readily institutionally identified as transitory, with $\rho_{\text{Top}} \gg \rho_{\text{Bot}}$.⁵ The labelled portion is excluded from her realised income shock $(\Delta e_t)^2$ before the σ_ψ^2 update; only the unlabelled residual feeds the belief rule, where she still has to attribute the residual between permanent and unlabelled transitory.

Mechanism

Given these primitives, we show how each of the three belief parameters arises.

Persistence μ_g . Bigger ω^2 at Bot makes her squared-shock signal noisier. A steady-state Kalman filter delivers a smaller gain μ_g , generating slower belief updating and the higher persistence observed at Bot.

Transitory prior $\sigma_{\theta, \text{perc}, g}^2$. The same noise pollutes her sample lag-1 autocovariance estimator of σ_θ^2 , making it less informative. A Bayesian agent therefore shrinks her noisy estimate toward

³US household income volatility is highest at the bottom of the distribution (Hardy and Ziliak, 2014), where it is driven primarily by hours, gig, and irregular tip income (Morduch and Schneider, 2017).

⁴Recession-driven employment and earnings losses fall disproportionately on low-education workers, with the gradient stable across US business cycles (Hoyne et al., 2012).

⁵Bonus, commission, and piecework pay are heavily concentrated at the top of the US wage distribution (Lemieux et al., 2009); consistent with this, positive earnings shocks at high incomes are predominantly transitory while at low incomes they are persistent (Guvenen et al., 2021).

a high prior $\bar{\sigma}_\theta^2$, with the strongest shrinkage at Bot, delivering her inflated $\sigma_{\theta,\text{perc}}^2$.

Attribution share λ_g . The agent's belief responds only to the unlabelled $(1 - \rho_g)$ share of $(\Delta e_t)^2$, while the regression (4) uses the full $(\Delta e_t)^2$ as the shock regressor. The labelled portion therefore adds variance to the regressor but does not move her belief, so the slope is mechanically diluted by the labelled variance share. Under joint Gaussianity (*derivation in Appendix C*) this gives the closed form

$$\lambda_g = \frac{\sigma_\psi^2(\sigma_\psi^2 + 2(1 - \rho_g)\sigma_\theta^2)}{(\sigma_\psi^2 + 2\sigma_\theta^2)^2}, \quad (6)$$

monotone-decreasing in ρ_g : Bot's small ρ therefore gives her a higher λ than Top's large ρ .

9 Embedding in a heterogeneous-agent model

To quantify the cross-tercile gradients of Section 7 and microfounded in Section 8, I embed the belief structure into an Aiyagari model under exogenous labour.

Setup

The wage process follows the Section 4 decomposition $e_{i,t} = p_{i,t} + \theta_{i,t}$ with $p_{i,t} = p_{i,t-1} + \psi_{i,t}$, primitives $(\sigma_\psi^2, \sigma_\theta^2)$ calibrated to SIPP. The two primitives enter at distinct points:

Paycheck noise ω_g^2 . The agent observes her paycheck change with labour-supply noise:

$$\Delta y_{i,t} = \Delta e_{i,t} + \nu_{i,t}, \quad \nu_{i,t} \sim \mathcal{N}(0, \omega_g^2).$$

Bot's higher ω^2 generates slower belief updating (lower μ_g) and an inflated transitory prior $\sigma_{\theta,\text{perc},g}^2$.

Labelled share ρ_g . A fraction ρ_g of each transitory shock arrives institutionally identified. The agent removes this share from $(\Delta y_t)^2$ and updates her permanent-variance belief $\hat{\sigma}_{\psi,t}^2$ on the unlabelled residual via the belief rule (2). Bot's smaller ρ_g lifts the attribution share λ_g .

Bellman

The agent solves

$$V_g(a, y, \hat{\sigma}_\psi^2) = \max_c \left\{ u(c) + \beta \mathbb{E}[V_g(a', y', \hat{\sigma}_{\psi'}^2) \mid y, \hat{\sigma}_\psi^2] \right\}, \quad \text{s.t. } a' = Ra + y - c, \quad a' \geq \underline{a}.$$

The belief state $\hat{\sigma}_\psi^2$ enters through the conditional expectation: a higher belief raises perceived variance and lifts saving. The expectation is taken under the agent's perceived process, not the realised one; the realised shock then drives the next paycheck Δy , which updates her belief.

Calibration

Given the SIPP wage primitives $(\sigma_\psi^2, \sigma_\theta^2)_g$, the structural parameters $(\omega_g^2, \rho_g, \bar{\sigma}_\theta^2)$ are picked via simulated method of moments (SMM) to match the headline-regression moments, following Bellifemine et al. (2025). Seven parameters (three ω_g^2 , three ρ_g , one $\bar{\sigma}_\theta^2$) against nine moments

(three each of $\hat{\mu}_g$, $\hat{\lambda}_g$, $\hat{\sigma}_{\theta, \text{perc}, g}^2$) leaves two over-identifying restrictions.

10 Conjectured macroeconomic implications

I conjecture the model of Section 9 generates two layers of under-prediction relative to a FIRE calibration.

Level. Bot perceives substantially more wage risk than Mid/Top, driven mainly by her inflated transitory prior. Her precautionary saving should run higher and her MPC out of transitory income lower than any FIRE calibration delivers. Under FIRE perceived risk is roughly equal across terciles, so the precautionary channel contributes no cross-tercile gradient and any MPC dispersion comes from liquidity constraints, not beliefs. The belief wedges therefore open a new gradient at Bot, partially offsetting the standard hand-to-mouth channel.

Dynamics. All three terciles' perceived variance responds positively to lagged wage shocks, with Bot's attribution share roughly twice Mid's and Top's. Under FIRE beliefs are static, so consumption responds only through current-income channels. The belief wedges therefore open a dynamic channel where realised shocks revise future-risk beliefs, with revisions concentrated at Bot.

Two illustrative examples. After a negative aggregate shock, Bot's perceived risk rises most sharply, lifting her precautionary motive and prolonging the consumption slump at the bottom. After a positive shock the sticky-belief side dominates: Bot's slow updating keeps her perceived risk elevated, depressing her MPC throughout the recovery. In both cases the belief wedge pushes against Bot's standard hand-to-mouth role, with the bottom acting as a drag on the cycle rather than amplifying it.

11 Next steps

- **Coefficient transformations.** The estimated $\hat{\lambda}_g$ in Section 6 sit roughly an order of magnitude below the literature, plausibly due to errors-in-variables bias from the cell-mean shock proxy. Im thinking about reporting cross-tercile ratios (Bot/Top) or applying a linear rescaling to absorb the uniform proxy bias and yield more comparable quantitative model magnitudes.
- **Asymmetric shocks.** Split the lagged squared shock into positive and negative components and re-estimate, testing whether perceived risk responds asymmetrically to upside vs downside wage news.
- **Conditional on liquid wealth.** Re-do the tercile split conditional on liquid wealth: low-income high-wealth households may update beliefs and respond to shocks differently from low-income low-wealth households, even within the same income tercile.
- **Full GE with endogenous labour.** Embed the calibrated belief structure in a general-equilibrium model with endogenous labour supply to recover consumption, labour-supply,

and portfolio responses. Coibion et al. (2023) is the inflation analogue (RCT pass-through from belief shocks to household actions), which finds significant effects in the aforementioned real variables. I would want to conclude this paper by comparing my quantitatively established responses in consumption, labour supply, and portfolio choice with these values.

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A From EWMA to observable AR(1)

Two ingredients. The EWMA on the agent's permanent-variance belief (2):

$$\hat{\sigma}_{\psi,t}^2 = (1 - \mu) \hat{\sigma}_{\psi,t-1}^2 + \mu \lambda (\Delta e_{t-1})^2,$$

and the level identity (3):

$$\bar{\sigma}_{\text{perc},t}^2 = 2\sigma_{\theta,\text{perc}}^2 + 12\hat{\sigma}_{\psi,t}^2.$$

Step 1. Solve the level identity for the belief:

$$\hat{\sigma}_{\psi,t}^2 = \frac{\bar{\sigma}_{\text{perc},t}^2 - 2\sigma_{\theta,\text{perc}}^2}{12}.$$

Step 2. Substitute both $\hat{\sigma}_{\psi}^2$ terms into the EWMA and multiply by 12:

$$\bar{\sigma}_{\text{perc},t}^2 - 2\sigma_{\theta,\text{perc}}^2 = (1 - \mu) [\bar{\sigma}_{\text{perc},t-1}^2 - 2\sigma_{\theta,\text{perc}}^2] + 12\mu\lambda (\Delta e_{t-1})^2.$$

Step 3. Collect the intercept: $2\sigma_{\theta,\text{perc}}^2 - 2(1 - \mu)\sigma_{\theta,\text{perc}}^2 = 2\mu\sigma_{\theta,\text{perc}}^2$.

Step 4. Add tercile subscripts and a measurement-error term:

$$\bar{\sigma}_{\text{perc},i,t}^2 = 2\mu_g \sigma_{\theta,\text{perc},g}^2 + (1 - \mu_g) \bar{\sigma}_{\text{perc},i,t-1}^2 + 12\mu_g \lambda_g (\Delta e_{i,t-1})^2 + \varepsilon_{i,t}.$$

This is (4).

B 12-month FIRE variance

Rolling the monthly process $e_t = p_t + \theta_t$ forward H months.

Step 1. Split the H -period change into its permanent and transitory pieces:

$$e_{t+H} - e_t = \underbrace{(p_{t+H} - p_t)}_{\text{permanent}} + \underbrace{(\theta_{t+H} - \theta_t)}_{\text{transitory}}.$$

Step 2. Permanent piece. The random walk $p_t = p_{t-1} + \psi_t$ accumulates one fresh innovation each month:

$$p_{t+H} - p_t = \sum_{k=1}^H \psi_{t+k}, \quad \text{Var}(p_{t+H} - p_t) = H\sigma_{\psi}^2$$

(ψ 's are iid, so variances of the sum add).

Step 3. Transitory piece. θ is iid, so only the endpoints matter:

$$\text{Var}(\theta_{t+H} - \theta_t) = \text{Var}(\theta_{t+H}) + \text{Var}(\theta_t) = 2\sigma_{\theta}^2.$$

Step 4. ψ and θ are independent shock processes, so the cross-covariance is zero and the total variance is the sum:

$$\text{Var}(e_{t+H} - e_t) = H\sigma_\psi^2 + 2\sigma_\theta^2.$$

At $H = 12$: $12\sigma_\psi^2 + 2\sigma_\theta^2$.

C Closed form for λ_g under partial labelling

Setup. Split the transitory shock $\theta_t = \theta_t^{\text{lab}} + \theta_t^{\text{unlab}}$ into independent labelled and unlabelled pieces with variances $\rho_g\sigma_\theta^2$ and $(1 - \rho_g)\sigma_\theta^2$. Substituting into the wage shock $\Delta e_t = \psi_t + \theta_t - \theta_{t-1}$ and grouping the pieces by labelling status gives

$$\Delta e_t = \underbrace{\psi_t + \theta_t^{\text{unlab}} - \theta_{t-1}^{\text{unlab}}}_{u_t, \text{ unlabelled residual}} + \underbrace{\theta_t^{\text{lab}} - \theta_{t-1}^{\text{lab}}}_{l_t, \text{ labelled component}},$$

where u_t collects everything the agent cannot identify ex ante (the permanent shock plus unlabelled transitory) and l_t is what she can. Both are zero-mean Gaussians, independent of each other, with $V_u \equiv \text{Var}(u) = \sigma_\psi^2 + 2(1 - \rho_g)\sigma_\theta^2$ and $V_l \equiv \text{Var}(l) = 2\rho_g\sigma_\theta^2$. Note $V_u + V_l = \sigma_\psi^2 + 2\sigma_\theta^2$.

Step 1. The agent's belief responds only to u_t^2 . Steady-state Bayesian consistency $\mathbb{E}[\hat{\sigma}_\psi^2] = \sigma_\psi^2$ pins the proportionality of belief-to- u^2 at σ_ψ^2/V_u . The slope coefficient identified by the regression (4) is therefore

$$\lambda_g = \frac{\sigma_\psi^2}{V_u} \cdot \frac{\text{Cov}(u_t^2, (u_t + l_t)^2)}{\text{Var}((u_t + l_t)^2)}.$$

Step 2. Under joint Gaussianity with $u \perp l$, expanding $(u + l)^2 = u^2 + 2ul + l^2$ and noting that cross-terms vanish in the relevant moments gives

$$\text{Cov}(u^2, (u + l)^2) = \text{Var}(u^2) = 2V_u^2, \quad \text{Var}((u + l)^2) = 2(V_u + V_l)^2.$$

Step 3. Substituting and using $V_u + V_l = \sigma_\psi^2 + 2\sigma_\theta^2$,

$$\lambda_g = \frac{\sigma_\psi^2}{V_u} \cdot \frac{2V_u^2}{2(V_u + V_l)^2} = \frac{\sigma_\psi^2 V_u}{(V_u + V_l)^2} = \frac{\sigma_\psi^2 (\sigma_\psi^2 + 2(1 - \rho_g)\sigma_\theta^2)}{(\sigma_\psi^2 + 2\sigma_\theta^2)^2}.$$

This is (6).